2019 IEEE 16th International Conference on Rehabilitation Robotics (ICORR)
Toronto, Canada, June 24-28, 2019

A Myoelectric Computer Interface for Reducing Abnormal Muscle Activations after Spinal Cord Injury

Fabio Rizzoglio*, Francesca Sciandra*, Elisa Galofaro, Luca Losio, Elisabetta Quinland, Clara Leoncini, Antonino Massone, F.A. Mussa-Ivaldi and Maura Casadio

***Abstract*— Myoelectric Computer Interfaces (MCIs) are a viable option to promote the recovery of movements following spinal cord injury (SCI), stroke, or other neurological disorders that impair motor functions. We developed and tested a MCI interface with the goal of reducing abnormal muscular activations due to compensatory strategies or undesired co-contraction after SCI. The interface mapped surface electromyographic signals (sEMG) into the movement of a cursor on a computer monitor. First, we aimed to reduce the co-activation of muscles pairs: the activation of two muscles controlled orthogonal directions of the cursor movements. Furthermore, to decrease the undesired concurrent activation of a third muscle, we modulated the visual feedback related to the position of the cursor on the screen based on the activation of this muscle. We tested the interface with six unimpaired and two SCI participants. Participants were able to decrease the activity of the targeted muscle when it was associated with the visual feedback of the cursor, but, interestingly, after training, its activity increased again. As for the SCI participants, one successfully decreased the co-activation of arm muscles, while the other successfully improved the selective activation of leg muscles. This is a first proof of concept that people with SCI can acquire, through the proposed MCI, a greater awareness of their muscular activity, reducing abnormal muscle simultaneous activations.**

I. Introduction

In the immediate aftermath of spinal cord injury, spinal shock causes a hyporeflexia condition that lead to a flaccid paralysis which involves the loss of muscle tone [1], [2]. Generally, within a few days to a few weeks from the injury, there is a transition to a hyperreflexia condition that might ultimately lead to spasticity [3], [4], [5]. Depending on the level and the severity of the lesion it is possible to predict the likelihood of developing spasticity. For example, for individuals with cervical SCI, the 93% of subjects with a complete injury classified as American Spinal Injury Association (ASIA) 'A' are likely to develop spasticity. This is also the case for 78% of subjects classified as ASIA 'B' to 'D' [6], [7]. Among the consequences of spasticity, there are changes in the reciprocal Ia inhibition [8], [9], that could determine the increase during voluntary contractions of undesired co-activations of antagonist muscles (i.e. increase in reciprocal facilitation [10]), and also the activation of muscles belonging to increasingly distant districts, even in the contralateral side [11]. Thus, spasticity affects the ability to perform selective movements and coordinate joints motion, resulting in impaired performance in functional tasks. Moreover, the difficulty of performing a specific movement typically lead SCI individuals to recruit other muscles to compensate for those originally involved in that movement [12], activating and strengthening undesirable muscle activation patterns. Therefore, it is essential for spinal cord injured individuals to reduce abnormal muscle co-activations within a rehabilitation program aimed at recovery in activities of daily living (ADL).

Myoelectric Computer Interfaces are an effective rehabilitative tool to address these problems. MCIs convert muscle activity, recorded by surface electromyography (sEMG), into control signals operating external devices, including computers. MCIs are frequently used in the prosthetic field, but they have been also used in motor rehabilitation protocols for reducing spasticity in stroke survivors [13], [14]. One of the factors that have limited the success of MCIs in rehabilitation is their prevalent use for providing mere biofeedback [15]. In fact, thinking which muscle to activate and trying to voluntary decrease the activity of the others could increment awareness of individual muscle activity, but it is difficult for people suffering from neurological conditions. We hypothesize that tasks where the goal of decoupling muscle activity is not rendered explicitly, could lead to better results. In this case participants will be motivated and engaged in solving a task aiming at a different goal, implicitly changing their muscle strategies. For example, unimpaired individuals are able to alter their motion strategies [16] or muscle activation patterns [17] pursuing the goal of optimizing the availability of visual information. Therefore, the manipulation of visual feedback could be a key element for MCIs when aiming at implicitly training muscle decoupling. This study aims at providing a first proof of concept for the feasibility of a training protocol based on a MCI aiming at changing muscle activation patterns. Specifically, the long-term goal of the proposed MCI is to reduce undesired muscle co-activations after spinal cord injury by including in the rehabilitative program, in addition to biofeedback of muscle activity, a manipulation of the visual feedback to enforce the desired

*equal contribution

Research supported by the Marie Curie Integration Grant (FP7-PEOPLE-2012-CIG-334201), the Ministry of Science and Technology, Israel (Joint Israel-Italy lab in Biorobotics "Artificial somatosensation for humans and humanoids"), the NIDILRR Grant 90REGE0005-01, the NICHHD Grant 5R01HD072080 and the NIBIB Grant R01-EB024058

F.R., F.S, E.G. M.C are with the University of Genoa, 16145 Genova, Italy (fabio.rizzoglio@edu.unige.it, francesca.sciandra94@gmail.com, elisa.galofaro@edu.unige.it, maura.casadio@unige.it)

L.L., E.Q., C.L., A.M. are with the S.C. Unità Spinale Unipolare, Santa Corona Hospital, ASL2 Savonese, Pietra Ligure, 17027 Savona, Italy (a.massone@asl2.liguria.it)

F.A.MI, F.R. are with Northwestern University and the Shirley Ryan Ability Lab, Chicago, IL, 60611, USA (sandro@northwestern.edu)

activity changes.

II. Methods

A. Participants

Six young unimpaired individuals (5 males and 1 female, age 30.50mean±12.63std years) participated in this study. They did not have any evidence or known history of postural, skeletal or neurological disorders and exhibited normal joint range of motion and muscle strength. The interface was also tested on two cervical SCI participants: SCI 1 (level of lesion C5, ASIA B age 44, female), SCI 2 (level of lesion C3, ASIA C age 19, male).

The study conformed to the Declaration of Helsinki and was approved by the local ethical committee (Comitato Etico Regionale - regione Liguria - 201REG2014). All participants signed an informed consent to participate in the study and to the analysis of their data for research purpose.

B. Experimental Set-up

Surface electromyographic (sEMG) signals were digitally recorded using FREEEMG 1000 system (BTS Bioengineering Corp., Quincy, MA, USA). The sampling frequency was 1 KHz. For the unimpaired participants, we recorded the activations of biceps brachii, lateral triceps, and lateral deltoid from the right side of the body. The same muscles were recorded for the SCI 1, but in this case we tested bilaterally. With the other SCI participant (SCI 2) we focused on the lower limb muscles, recording the left rectus femoris as well as the left and right semitendinosus. Each raw signal was preprocessed online to obtain the EMG envelopes used for controlling the position or the visibility of the cursor. Specifically, a bandpass ($fc = 30 - 450$ Hz) 2nd order IIR biquadratic Butterworth filter was used to remove high frequency noise and low frequency motion artifacts. Afterwards, the signal was rectified and a low-pass filter ($fc = 2$Hz) was applied.

Each session started with a calibration phase where users were asked to perform an isometric Maximum Voluntary Contraction (MVC) task on each muscle for three times. The maximum value obtained in the three repetitions was then saved for further defining the level of activation required in each session. This calibration was followed by a rest phase in which users were asked to fully relax their muscles for 10s. The signals recorded during this resting period were used to subtract from each muscle possible offsets that, while not corresponding to variation on the muscle activations, could have affected the online control of the cursor.

The interface mapped the sEMG from a pair of muscles ($m1$ and $m2$) to two cursor coordinates on the monitor as:

$$\begin{aligned} x &= \frac{sEMG_{m1}}{c_1} * d \\ y &= \frac{sEMG_{m2}}{c_2} * d \end{aligned} \tag{1}$$

where sEMG indicates the actual value of the EMG envelope, d is the distance (7.2 cm) between the peripheral and the home target in the monitor workspace (see Fig. 1) and c_1 and c_2 are the scaling values obtained from the MVC for $m1$ and $m2$ respectively at the beginning of each session. Thus $m1$ controlled the horizontal movement of the cursor while $m2$ controlled the vertical movement. A combination of $m1$ and $m2$ moved the cursor through the entire workspace.

For unimpaired participants and SCI 1, we recorded the activity of a third muscle ($m3$) and we controlled the cursor visibility proportionally to its activation. The visibility of the cursor varied from transparent (0%) to maximally visible (100%) as follow:

$$visibilty = \begin{cases} 100\%, & m_3 \leq 5\%_{MVC} \\ 75\%, & 5\%_{MVC} \leq m_3 < 10\%_{MVC} \\ 50\%, & 10\%_{MVC} \leq m_3 < 15\%_{MVC} \\ 25\%, & 15\%_{MVC} \leq m_3 < 20\%_{MVC} \\ 0\%, & m_3 \geq 20\%_{MVC} \end{cases} \tag{2}$$

Cursor visibility encoded the additional goal of reducing $m3$ activity while activating $m1$ and $m2$. The interface was developed in C# as a Windows Form Application in Visual Studio 2017, with a targeted framework of .NET 4.5.2. The graphical part of the MCI was developed using OpenTK, a C# graphic library that provided access to graphic tools contained in OpenGL. Graphics refresh rate was set to 50 Hz.

C. Experimental Protocol

With the proposed MCI, we aimed to address whether individuals with SCI were able to:

1) Decouple a pair of muscles ($m1$ and $m2$) that they were co-contracting abnormally;
2) Decrease the activity of a muscle ($m3$) that they were concurrently activating with $m1$ and $m2$ due to abnormal strategies.

To validate the interface and identify potential issues that could impact the SCI participants, we tested the MCI on a group of unimpaired participants. The validation was directed at assessing whether unimpaired participants would be able of learning with practice to control the MCI with the independent contraction of two antagonist muscles ($m1$ and $m2$). More importantly, we also verified whether the interface would enforce changes in the activity of a muscle ($m3$) that tended to be naturally coactivated with $m1$ and $m2$.

The protocol was based on the execution of a reaching task. Participants were seated in front of a 15.6" PC monitor at a distance of 1m. The cursor and the targets were shown on the monitor. The participants were asked to reach the targets with straight movements of the cursor. Targets in the workspace were placed as shown in Fig. 1.

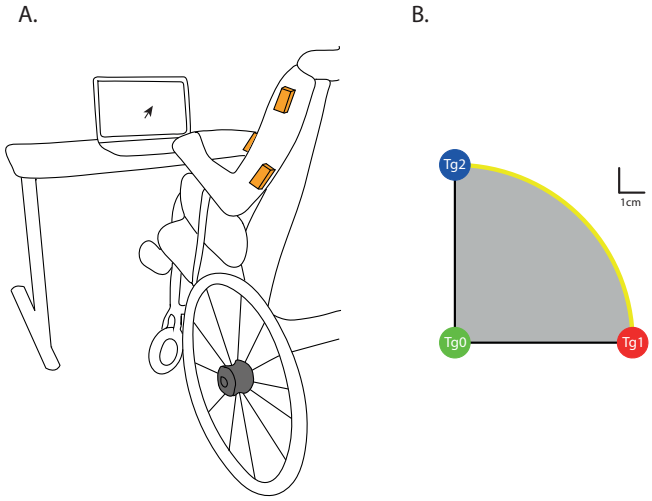

Fig. 1. Panel A: setup for reaching task. Panel B: targets positions on PC monitor. Home Target (green - Tg0) corresponded to the position of the cursor during rest. Target 1 (red - Tg1) was reached by contracting only $m1$. Target 2 (blue - Tg2) was reached by contracting only $m2$.

The protocol was divided in three phases:

1) **Training 1**: this phase was introduced to let users learn to decouple $m1$ and $m2$. In this phase, the two peripheral targets had to be reached 10 times each in random order. A reaching movement was considered successful if the cursor remained inside the target for 0.3s. We used an adaptive protocol implicitly asking users to increase the activation of one muscle without a concurrent increase of the other. In fact, the muscular activity needed to reach the target with the cursor gradually increased throughout the training. $c1$ and $c2$ cursor scaling values described in equation (1) were initially set at 20% of the corresponding muscle MVC. The subsequent thresholds were respectively set at 25 - 30 - 35 - 40% of the MVC. Note that when the cursor was inside a target the activity of the muscle controlling the opposite direction was close to 0. To increase the threshold for one target, it was necessary to consecutively reach it for three times in less than 60s. Instead, if a target was not reached within 60s, the corresponding threshold was lowered to the previous level.
2) **Training 2**: this phase was introduced to let users learn to decrease the activity of $m3$. The protocol was the same as described in Training 1, but here we introduced a variability in the visual feedback given to the users: the more active was the $m3$ muscle, the less visible was the cursor. We hypothesized that users would decrease this muscle activation to maximize the availability of visual feedback.
3) **Test**: during this phase users had to reach 18 times each of the two peripheral targets. A reaching movement ended when the cursor remained inside the target for 0.3s or when it remained on any point lying on the arc joining Tg1 and Tg2 (yellow curve shown in Fig. 1) for 1s (being, therefore, at 7.2 cm distance from Tg0). At the end of each movement, we provided users with qualitative feedback of their performance by showing the value of their score (see next section for further details). In this phase, $c1$ and $c2$ were set at 40% of the MVC of the corresponding muscle.

D. Unimpaired participants - Sessions

To validate our approach, we performed a preliminary test on a group of unimpaired participants. Their protocol consisted of 5 phases presented in the order: Test (BASE), Training 1, Test (MID), Training 2, Test (FINAL). They used their right lateral triceps ($m1$) and biceps brachii ($m2$) to control the cursor as well as in Training 2 the right lateral deltoid as $m3$ muscle to modify cursor visibility.

E. SCI participants - Sessions

As for the SCI participants, they were involved in a personalized multisession training, where they were asked to perform isometric contractions of different muscles to move the cursor. The duration of each session was kept short to limit the influence of muscle fatigue on performance.

Specifically, SCI 1 participated in seven rehabilitative sessions for the left arm and twelve for the right arm, with a frequency of three sessions per week. Biceps brachii was used as $m1$, lateral triceps as $m2$ and lateral deltoid as $m3$. Table 1 summarizes the rehabilitative sessions for SCI 1. Each Test session consisted of 10 trials, 5 for each target.

TABLE I

REHABILITATIVE SESSIONS FOR SCI 1

Left (contralateral) muscles		Right (ipsilateral) muscles	
#sess		#sess	
1	Baseline Test	1	Baseline Test
2-3	Test - Training 1 - Test	2-3	Test - Training 1
4-6	Test - Training 2 - Test	4-6	Test - Training 2
7	Final Test	7	Mid Test
		8-11	Test - Training 2
		12	Final Test

Conversely, the protocol used for SCI 2 had to be strongly customized due to his critical clinical conditions. In particular, he presented severe issues at the respiratory level and high spasticity that sometimes led to the stiffening of the whole body. We, therefore, decided to have him performing only the Test and using it as practice for decreasing co-contraction and increasing awareness of the lower limb muscles. This phase differed from that used for SCI 1 as we increased the number of repetitions for each target from 5 to 18. Three Test sessions were performed for the reduction of left leg muscles co-activation (left semitendinosus as $m1$ and left rectus femoris as $m2$) and four Test sessions for the reduction of pathological co-activation between right ($m1$) and left ($m2$) semitendinosus, with an average frequency of two weekly sessions.

TABLE II

REHABILITATIVE SESSIONS FOR SCI 2

Left leg muscles		Right-left semitendinosus	
#sess		#sess	
1-3	Test	1-4	Test

F. Data Analysis

Three different parameters were chosen to evaluate user's performance:

- **Movement Time (MT)**: time to reach the peripheral targets after leaving the home target;
- **Smoothness Index (NPK)**: expressed as the number of peaks (maxima) in the speed profile. We counted the peaks greater than 15% of the maximum speed in each trajectory. This is a measure of smoothness, based on how many cursor sub-movements were included in each reaching movement;
- **Score**: accuracy index reporting on how much the cursor position deviated from the ideal trajectory during a center-out reaching movement. During a trial, every 0.25s we computed the angle (θ) between the straight lines connecting cursor and the target to the starting position. Then, we computed the average of those values at the end of the trial. When $\theta = 90°$ the cursor was totally deviating from the ideal trajectory; thus, in order to have this metric ranging from 0 indicating the worst performance to 100 indicating the best performance, we defined the score as:

$$score = 100 * (1 - \frac{\theta}{90}) \tag{3}$$

A score of 100 corresponds to the participant activating only $m1(m2)$, while maintaining $m2(m1)$ below its resting value. The score was shown to the user at the end of each trial.

To verify how users were varying the activation of $m3$ we computed:

- **Activation of $m3$ (AM3)**: % of time over the entire trial duration in which $m3$ was higher than 5% of its MVC.

To test how the indicators related to the motor strategies changed in unimpaired participants we ran repeated measures ANOVA with two within-subjects factors: pratice (1-3: BASE, MID, FINAL for MT, NPK and Score; MID, TRAIN2, FINAL for AM3) and targets (1-2: Tg1, Tg2). A post-hoc analysis (Fisher's LSD test) was used to verify statistically significant differences among factors after repeated measures ANOVA. Significance was asserted when $p < 0.05$. The statistical analysis was computed within Statsoft environment (Statistica software 10.0, Statsoft TULSA, USA).

A single subject design was chosen for SCI participants. Data were analyzed with a visual inspection focusing on the time trend of the measurements and their variability. To verify whether the difference in average between two sets of observations made on the same patient was different from 0, we ran a Wilcoxon rank sum test.

III. Results

A. Unimpaired participants

Repeated measures ANOVA ran on unimpaired participants revealed an effect of training on movement time (MT: $F(2,10) = 7.80, p = 0.009$) and smoothness (NPK: $F(2,10) = 11.42, p = 0.003$). No significant effects of target positions or interactions with practice were found. Participants became significantly faster and smoother from baseline to the final session (BASE vs FINAL - MT: $p = 0.006$, NPK: $p = 0.002$). They also improved the score (BASE vs FINAL - $p = 0.037$). There was significant learning across the first training block (BASE vs MID - MT: $p = 0.007$; NPK: $p = 0.003$). Additional training with cursor visibility modulation did not lead to further improvement on any performance metric (MID vs FINAL).

As for the activation of the lateral deltoid ($m3$), we found a significant effect of practice (AM3: $F(2,10) = 12.11, p = 0.002$). No significant effects of targets position or interactions with practice were found. Specifically, participants significantly decreased lateral deltoid activation during Training 2 with respect to both Mid Test (MID vs TRAIN2 - $p = 0.001$) and Final Test (TRAIN2 vs FINAL - $p = 0.002$). However, no differences were found between activations during Mid and Final Test (MID vs FINAL).

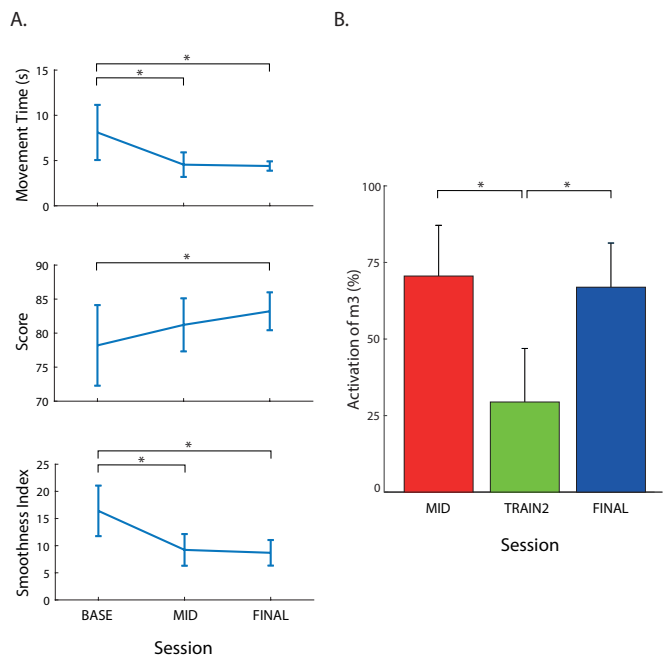

Fig. 2. Panel A: performance metrics values (from top to bottom: MT, Score and NPK) for unimpaired participants during the three sessions of test (BASE, MID and FINAL). Panel B: activation of lateral deltoid ($m3$) during mid test, second session of training (when cursor visibility modulation was active) and final test (red bar indicates MID test, green bar indicates TRAIN2 and blue bar indicates FINAL test). Mean across targets and participants is plotted with standard error.

B. First SCI participant

SCI 1 participated in 7 sessions with the left upper arm. She was able to generally improve the performance from the

first to the last session. In particular, the movements became faster (MT: $p_w < 0.001$ for Tg1) and smoother (NPK: $p_w = 0.008$ for Tg2). Concurrently, Score significantly increased when reaching Tg2 ($p_w < 0.001$) but nor for Tg1 ($p_w = 0.236$), while AM3 significantly decreased for both Tg1 ($p_w = 0.001$) and Tg2 ($p_w < 0.001$).

She participated also in 12 sessions with the right upper arm. There was a significant improvement between Baseline and Final Test for MT ($p_w < 0.001$ for both Tg1 and Tg2) and NPK ($p_w = 0.047$ for Tg1, $p_w < 0.001$ for Tg2). Score significantly increased as well, but only for Tg1 ($p_w = 0.002$). She was able to decrease the right lateral deltoid activity at the end of the rehabilitative session, with a significant difference from the initial session for Tg2 ($p_w < 0.001$), but not for Tg1.

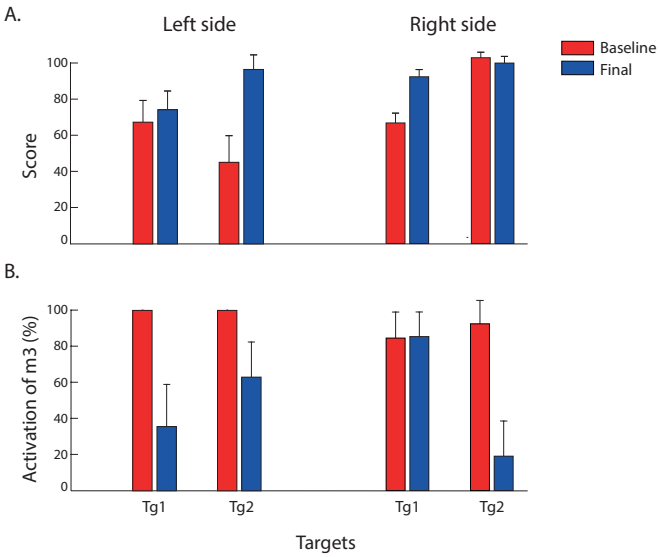

Fig. 3. Panel A: (Left) Score values for left muscles during Baseline (red) and Final (blue) test session and (right) score values for right muscles during Baseline (red) and Final (blue) test session when reaching Tg1 and Tg2. Panel B: (Left) Activation of left lateral deltoid during Baseline (red) and Final (blue) test session and (right) activation of right lateral deltoid during Baseline (red) and Final (blue) test session when reaching Tg1 and Tg2. Mean across repetitions is plotted with standard error.

C. Second SCI participant

SCI 2 participated in three sessions for reducing the co-contraction between left rectus femoris and semitendinosus. When reaching Tg2 (i.e. when he had to primarily activate the left rectus femoris muscle), he increased the score ($p_w < 0.001$) and moved faster (MT: $p_w = 0.022$) and smoother (NPK: $p_w = 0.001$).

He was also involved in four sessions for reducing the pathological co-activation between left and right semitendinosus. From first to last session, he moved faster (MT: $p_w = 0.05$) and smoother (NPK: $p_w = 0.148$) when moving the cursor towards Tg2, indicating an improved selective activation of the left semitendinosus. Score improved as well ($p_w < 0.001$).

Unfortunately in both cases these positive changes were not observed for Tg1 (i.e. respectively when he had to primarily contract the left semitendinosus over the left rectus femoris and the right semitendinosus over the left one).

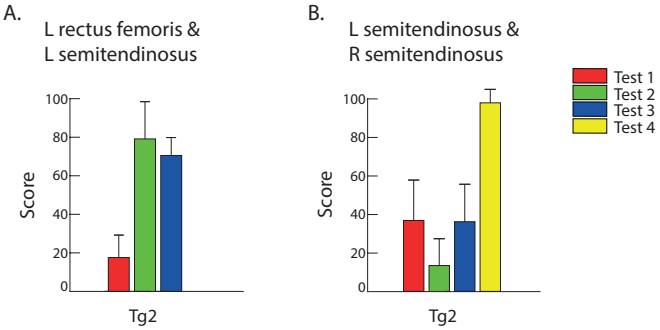

Fig. 4. Panel A: Score values during the three sessions of test for reducing co-activation between left rectus femoris and semitendinosus when reaching Tg2 (first session is plotted in red, second session in green and third session in blue). Panel B: Score values during the four sessions of test for reducing co-activations between left and right semitendinosus when reaching Tg2 (first session is plotted in red, second session in green, third session in blue and fourth session in yellow). Mean across repetitions is plotted with standard error.

IV. Conclusions

With the proposed MCI we aimed at reducing abnormal muscle simultaneous activations in people with SCI. First, we validated our interface on a group of unimpaired users and then we tested it on two SCI participants by targeting different muscles depending on individual needs.

The results on unimpaired users showed that they improved their performances with practice. Moreover, they decreased the activity of the lateral deltoid but only when cursor visibility was linked with its activation. This confirms the effectiveness of manipulating visual feedback when developing rehabilitation exercises: we think that the retention of this effect may be obtained by increasing the training time. However, this critical conclusion requires additional, more prolonged experiments.

As for the first SCI participant, we tested both the ipsilateral and the contralateral side of the lesion. For the latter, we found a significant improvement from first to last session for all metrics, meaning that she was able to decrease the co-activations between lateral triceps and biceps brachii. For the ipsilateral side, the score did not improve when she had to generate an elbow flexor torque without activating the triceps: this was due to the fact that score was already close to its maximum during baseline. We noticed that the activity of the lateral deltoid by compensatory strategy decreased as well with practice, as she successfully reached a minimum activation during final test, except when she had to primarily contract the ipsilateral triceps.

As for the second SCI participant, we would like to first point out that his was a case of high clinical impairment. We therefore decided to test the interface on the lower limbs, as excessive spasticity affected the upper limb level. We obtained mixed results, as the selective activations of left rectus femoris and semitendinosus improved only when generating an extensor torque of the knee without activating the corresponding antagonist muscle (i.e semitendinosus).

This is consistent with [18], as extensor activity generally outlasts flexor activity in SCI due to extensor spasms, that might be triggered by proprioceptors in the hip. Similarly, he reduced the irradiation between left and right semitendinosus only when moving towards the target that required to activate only the left semitendinosus.

Therefore, we concluded that our interface could be used as a valid rehabilitative tool when trying to reduce dysfunctional muscle activations resulting from neurological disorders. We expect that these results will be confirmed in experiments with a larger population of SCI participants.

REFERENCES

[1] C. S. Sherrington, "Flexion-reflex of the limb, crossed extension-reflex, and reflex stepping and standing," *The Journal of physiology*, vol. 40, no. 1-2, pp. 28–121, 1910.

[2] J. Ditunno, J. Little, A. Tessler, and A. Burns, "Spinal shock revisited: a four-phase model," *Spinal cord*, vol. 42, no. 7, p. 383, 2004.

[3] V. Dietz, "Spastic movement disorder," *Spinal cord*, vol. 38, no. 7, p. 389, 2000.

[4] J. W. Lance, "The control of muscle tone, reflexes, and movement robert wartenbeg lecture," *Neurology*, vol. 30, no. 12, pp. 1303–1303, 1980.

[5] M. M. Adams and A. L. Hicks, "Spasticity after spinal cord injury," *Spinal cord*, vol. 43, no. 10, p. 577, 2005.

[6] C. Sköld, R. Levi, and Å. Seiger, "Spasticity after traumatic spinal cord injury: nature, severity, and location," *Archives of physical medicine and rehabilitation*, vol. 80, no. 12, pp. 1548–1557, 1999.

[7] F. Maynard, R. Karunas *et al.*, "Epidemiology of spasticity following traumatic spinal cord injury." *Archives of physical medicine and rehabilitation*, vol. 71, no. 8, pp. 566–569, 1990.

[8] N. Yanagisawa, "Reciprocal reflex connections in motor disorders in man. in desmedt je (ed): Spinal and supraspinal mechanisms of voluntary motor control and locomotion," *Prog Clin Neurophysiol*, vol. 8, pp. 129–141, 1980.

[9] Y. Okuma, Y. Mizuno, and R. G. Lee, "Reciprocal ia inhibition in patients with asymmetric spinal spasticity," *Clinical neurophysiology*, vol. 113, no. 2, pp. 292–297, 2002.

[10] S. Cremoux, D. Amarantini, J. Tallet, F. Dal Maso, and E. Berton, "Increased antagonist muscle activity in cervical sci patients suggests altered reciprocal inhibition during elbow contractions," *Clinical Neurophysiology*, vol. 127, no. 1, pp. 629–634, 2016.

[11] J. Bonavita, M. Menarini, and P. Pillastrini, *La riabilitazione nelle mielolesioni*. Elsevier srl, 2004.

[12] K. S. Beekhuizen, "New perspectives on improving upper extremity function after spinal cord injury," *Journal of Neurologic Physical Therapy*, vol. 29, no. 3, pp. 157–162, 2005.

[13] R. Song, K.-y. Tong, X. Hu, L. Li *et al.*, "Assistive control system using continuous myoelectric signal in robot-aided arm training for patients after stroke," *IEEE transactions on neural systems and rehabilitation engineering*, vol. 16, no. 4, pp. 371–379, 2008.

[14] Z. A. Wright, W. Z. Rymer, and M. W. Slutzky, "Reducing abnormal muscle coactivation after stroke using a myoelectric-computer interface: a pilot study," *Neurorehabilitation and neural repair*, vol. 28, no. 5, pp. 443–451, 2014.

[15] O. M. Giggins, U. M. Persson, and B. Caulfield, "Biofeedback in rehabilitation," *Journal of neuroengineering and rehabilitation*, vol. 10, no. 1, p. 60, 2013.

[16] S.-H. Yeo, D. W. Franklin, and D. M. Wolpert, "When optimal feedback control is not enough: Feedforward strategies are required for optimal control with active sensing," *PLoS computational biology*, vol. 12, no. 12, p. e1005190, 2016.

[17] J. Mintz, D. De Santis, F. Rizzoglio, A. Farshchiansadegh, and F. A. Mussa-Ivaldi, "Designing visual feedback to reshape muscle coordination," in *International Conference on NeuroRehabilitation*. Springer, 2018, pp. 1034–1038.

[18] B. D. Schmit and E. N. Benz, "Extensor reflexes in human spinal cord injury: activation by hip proprioceptors," *Experimental brain research*, vol. 145, no. 4, pp. 520–527, 2002.